

Heart Disease Prediction using Supervised Machine Learning

A Comparative Study Across Four Hospitals

Presented by

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≈ 1/3

of global deaths

are caused by cardiovascular disease — many cases remain undiagnosed until a serious cardiac event occurs.

Our Approach

Binary classification of heart-disease risk

INPUT

13 clinical features per patient — age, sex, chest-pain type, blood pressure, cholesterol, max heart rate, ST depression, etc.

OUTPUT

1 = heart disease present / 0 = no disease

WHY IT MATTERS

An accurate, transparent screening model can support clinicians in triaging patients and prioritizing follow-up tests — especially in resource-limited settings.

918

Unique Patients

after de-duplication

4

Source Hospitals

Cleveland, Hungary, Switzerland, VA

13

Clinical Features

age, sex, cp, thalch, oldpeak, ...

55%

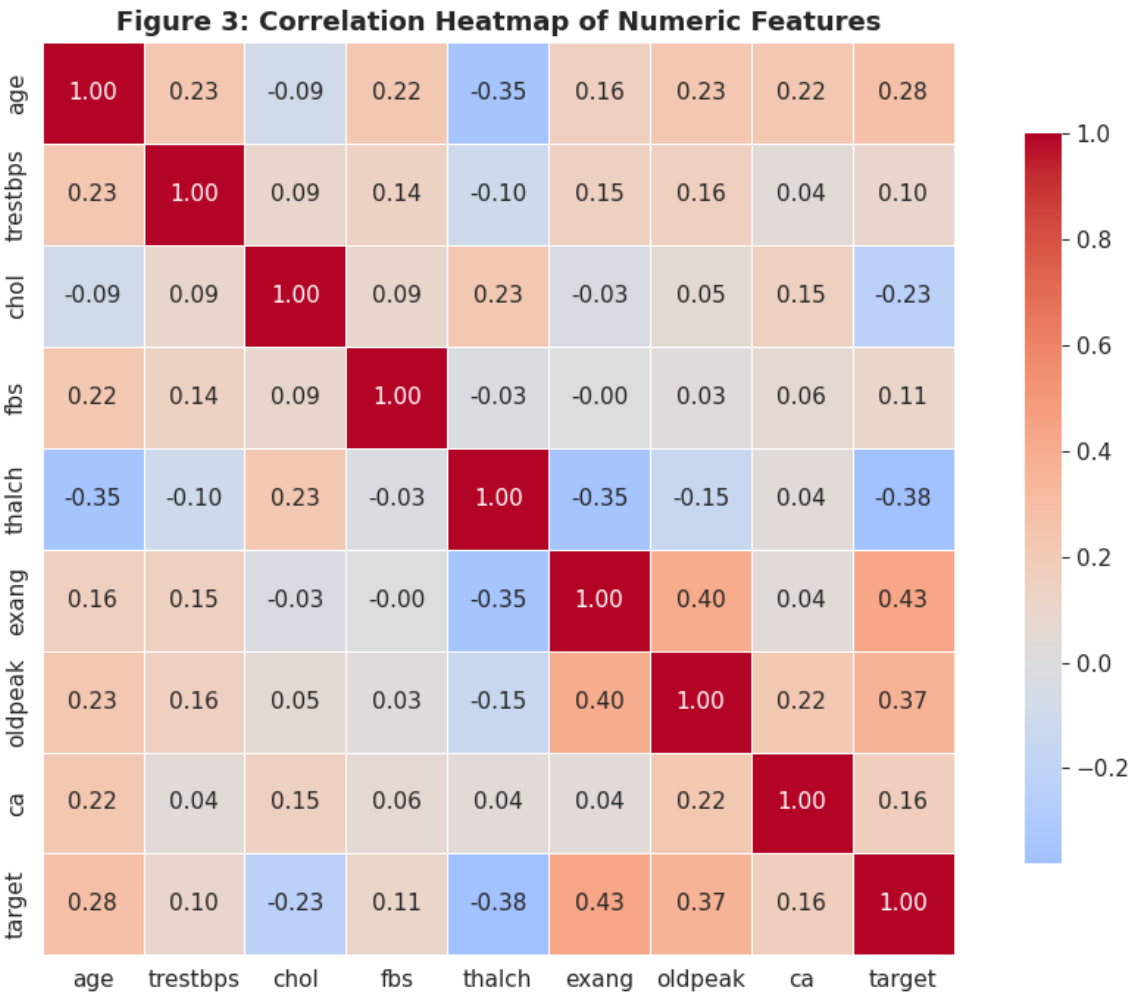
Disease Prevalence

binary target (mildly imbalanced)

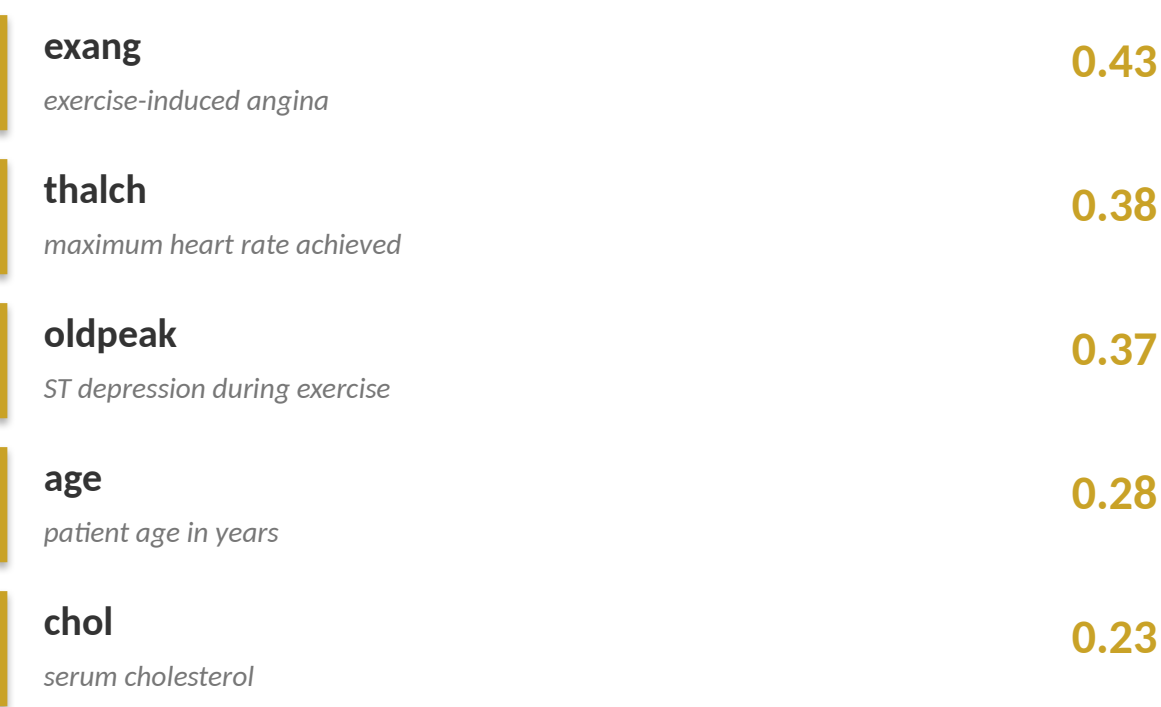
Key Preprocessing Steps

- ▶ Converted multi-class severity num (0–4) → binary target (num > 0).
- ▶ Handled heavy missingness (ca: 66.3%, thal: 52.7%) with median / mode imputation.
- ▶ One-hot encoded 6 categorical features → 21-dimensional feature space.
- ▶ Stratified 80 / 20 split (734 train / 184 test); StandardScaler fit on train only — no data leakage.

Correlation with Target



Top Predictors (numeric)



KEY INSIGHT

Disease prevalence varies widely by hospital — directly relevant for the fairness audit.

Methodology — Four Algorithms

Spanning four distinct algorithmic families

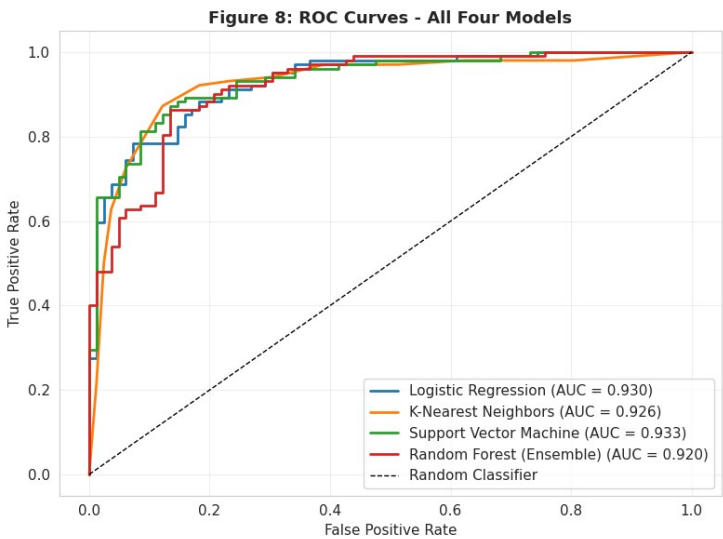
LINEAR	INSTANCE-BASED	MARGIN-BASED	ENSEMBLE
Logistic Regression Models log-odds of disease as a weighted sum of features. Provides interpretable coefficients valuable in clinical contexts. L2 regularization tuned via C.	K-Nearest Neighbors Assigns class by majority vote of the k closest patients in standardized feature space. Tuned k, distance metric, and weighting scheme. No global decision boundary.	Support Vector Machine Finds the maximum-margin hyperplane between classes. Linear and RBF kernels tested; C tunes margin vs. error trade-off. Strong on small clean datasets.	Random Forest Averages 200 decision trees trained on bootstrap samples. Reduces variance dramatically and produces feature-importance scores used to interpret predictions.

Results — Model Comparison

Test-set performance (n = 184)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8424	0.8288	0.9020	0.8638	0.9298
K-Nearest Neighbors ★	0.8750	0.8624	0.9216	0.8910	0.9258
Support Vector Machine	0.8533	0.8261	0.9314	0.8756	0.9328
Random Forest (Ensemble)	0.8533	0.8378	0.9118	0.8732	0.9199

ROC Curves



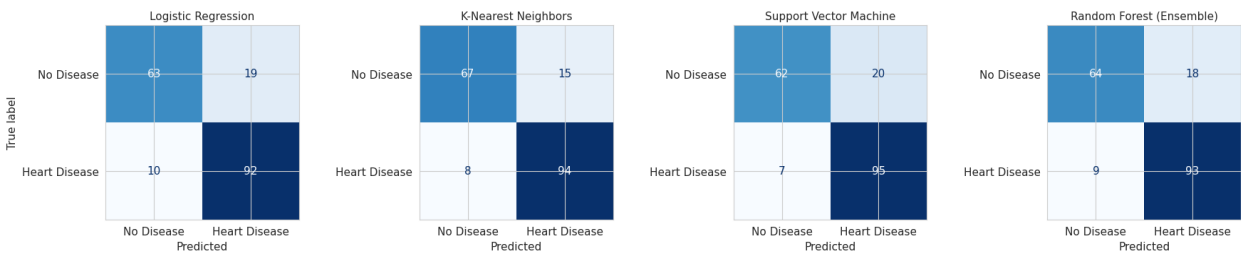
🏆 BEST MODEL

K-Nearest Neighbors

Accuracy: 87.5% · F1-Score: 0.891 · ROC-AUC: 0.926 · Recall: 92.2%

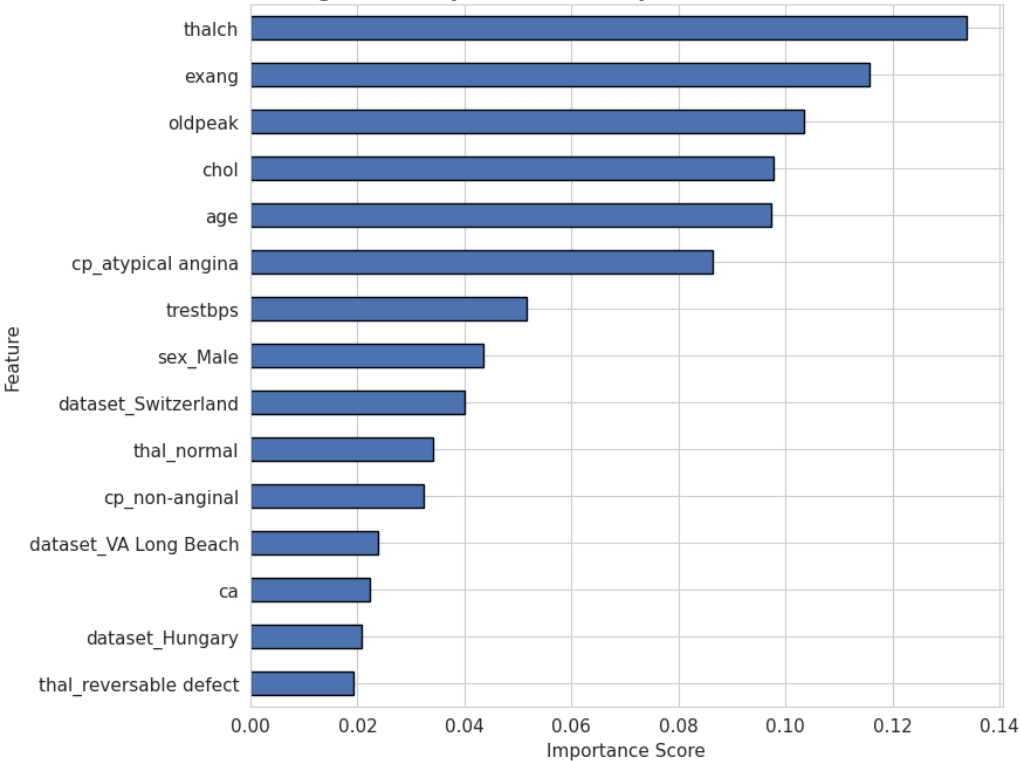
Confusion Matrices

Figure 7: Confusion Matrices - All Four Models



Top Features (Random Forest)

Figure 10: Top 15 Feature Importances - Random Forest



CLINICAL VALIDATION

Top predictors (thalch, exang, oldpeak) match established cardiology literature on coronary risk.

Why KNN Won

- ▶ After standardization, patients form compact clusters → ideal for distance-based methods.
- ▶ Random Forest slightly overfits the small (~918-row) dataset on the training folds.
- ▶ All four models cluster at ROC-AUC 0.92–0.93: preprocessing matters more than model choice.
- ▶ Logistic Regression is nearly as accurate AND interpretable — a strong clinical option.

Fairness Audit of KNN

Subgroup	n	Accuracy	Recall
Female	35	0.943	0.875
Male	149	0.859	0.926
Cleveland	55	0.873	0.923
Hungary	65	0.846	0.739
Switzerland	26	1.000	1.000
VA Long Beach	38	0.842	1.000

⚠ ETHICAL CONCERNS IDENTIFIED

Female patients (21% of data) get lower recall. Hungarian cohort recall drops to 73.9% due to missing ca/thal features. → Model must be a screening aid, never an autonomous decision-maker.

Conclusion

- ▶ Built a complete ML pipeline on UCI Heart Disease (918 patients, 4 hospitals).
- ▶ KNN achieved the best balance: 87.5% accuracy, F1 = 0.891, ROC-AUC = 0.926.
- ▶ Top predictors (thalch, exang, oldpeak) align with cardiology literature.
- ▶ Fairness audit revealed performance gaps by sex and hospital — disclosed honestly.

Future Work

- ▶ Try gradient-boosting (XGBoost, LightGBM) and stacking ensembles.
- ▶ Use SHAP values for per-patient explainability — essential for clinical adoption.
- ▶ Apply fairness-aware methods (reweighting, adversarial debiasing).
- ▶ Replace simple imputation with MICE / k-NN imputation for ca and thal.
- ▶ Validate on truly external clinical data, not just a held-out split.

TOOLS & LIBRARIES

Python · scikit-learn · pandas · numpy · matplotlib · seaborn · Google Colab

Thank You

Questions · Discussion · Feedback

KEY REFERENCES

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